

MATH 4790: Quiz 6

April 23, 2026

Name: _____

Score: _____/15

Instructions: Show all work and justify your answers. Only calculators approved for Exam FM are permitted. Credit will not be given to just guessing the correct answer.

Question 1.

Logo purchases an annuity that pays monthly amounts as follows:

- Payments of X at times $t = 18, 19, \dots, 35$
- Payments of $X + 500(t - 35)$ at times $t = 36, 37, \dots, 71$

where t is measured in months.

The annual effective interest rate is 10%.

The present value of all payments is 370,763.65.

Calculate X .

- (A) 3,450 (B) 3,980 (C) 4,250 (D) 4,430 (E) 5,210

Question 2.

Erin borrows X at an annual effective rate of 5%. If the principal and all accumulated interest is paid in one lump sum at the end of 20 years, \$1,000 more in interest would be paid than if the loan was repaid with level payments at the end of each of the first ten years.

Calculate X .

- (A) 389 (B) 540 (C) 704 (D) 736 (E) 954

Question 3.

A company has liabilities of 402.11 due at the end of each of the next three years.

The company will match the duration of its liabilities by investing a total of 1,000 in one-year and three-year zero-coupon bonds. The annual effective yield of both bonds is 10%.

Calculate the amount the company will invest in one-year bonds.

- (A) 366 (B) 402 (C) 442 (D) 500 (E) 532